Lab 3 practice: first-order DEs, all types

Zill §2.2 to 2.5. Prep for the Wed Sep 30 lab quiz. Closed book, about 10 minutes per problem. The equations are in no particular order and none of them is labelled. For each one, name the type and say why before you start solving; that is the skill this quiz tests. Some fit more than one type. Every solution was checked by substituting back (`lab3_check.py`).

Which type is it? Check in this order.

Type	Test	Method
Separable	$y' = g(x)h(y)$	$\int dy/h = \int g\,dx$; check $h(y) = 0$ for lost solutions
Linear	$y' + P(x)y = f(x)$	$\mu = e^{\int P\,dx}$, then $(\mu y)' = \mu f$
Exact	$M\,dx + N\,dy = 0$ with $M_y = N_x$	$F = \int M\,dx + g(y)$, set $F_y = N$; answer $F = C$
Not exact, $\mu(x)$	$(M_y - N_x)/N$ depends on x only	$\mu = e^{\int (M_y - N_x)/N\,dx}$
Not exact, $\mu(y)$	$(N_x - M_y)/M$ depends on y only	$\mu = e^{\int (N_x - M_y)/M\,dy}$
Homogeneous	$f(tx, ty) = f(x, y)$	$y = ux$, so $y' = u + xu'$; separable in u and x
Bernoulli	$y' + Py = fy^n$	$u = y^{1-n}$; linear in u
$f(Ax + By + C)$	y' depends only on $Ax + By + C$	$u = Ax + By + C$; separable

Finish. Apply initial conditions last. State the interval of validity and any singular solutions.

Problem 1

$x\frac{dy}{dx} + y = x^2y^2$

Solution: Bernoulli, $n = 2$

It looks linear, but the y^2 on the right rules that out. Divide by x: $y' + \frac{1}{x}y = xy^2$, a Bernoulli equation with $n = 2$.

Let $u = y^{1-2} = y^{-1}$, so $y = u^{-1}$ and $y' = -u^{-2}u'$. Substituting and multiplying through by $-u^2/x$:

$$u' - \frac{1}{x}u = -x$$

That is linear in u. The integrating factor is $e^{-\int dx/x} = 1/x$:

$$\left(\frac{u}{x}\right)' = -1 \quad \Longrightarrow \quad \frac{u}{x} = -x + C \quad \Longrightarrow \quad u = Cx - x^2$$

$$\boxed{y = \frac{1}{Cx - x^2}}$$

$y = 0$ also solves the equation, and no value of C gives it: it is a **singular solution**, lost when we divided by y^2.

Problem 2

$(2xy + \cos y)\,dx + (x^2 - x\sin y + 2y)\,dy = 0$

Solution: exact

$M = 2xy + \cos y$ and $N = x^2 - x\sin y + 2y$, so $M_y = 2x - \sin y = N_x$. Exact.

$$F = \int M\,dx = x^2y + x\cos y + g(y)$$

$$F_y = x^2 - x\sin y + g'(y) = N \quad \Longrightarrow \quad g'(y) = 2y,\ g = y^2$$

$$\boxed{x^2y + x\cos y + y^2 = C}$$

Problem 3

$\frac{dy}{dx} = (x + y + 1)^2$

Solution: substitution, $u = x + y + 1$

The right side is a function of $x + y + 1$ alone. Let $u = x + y + 1$, so $u' = 1 + y'$:

$$u' = 1 + u^2 \quad \Longrightarrow \quad \int \frac{du}{1+u^2} = \int dx \quad \Longrightarrow \quad \tan^{-1} u = x + C$$

$$\boxed{y = \tan(x + C) - x - 1}$$

$1 + u^2$ is never zero, so there are no constant or singular solutions to worry about.

Problem 4

$x\dfrac{dy}{dx} + 2y = \sin x, \qquad y(\pi) = 0$

Solution: linear IVP

Standard form: $y' + \frac{2}{x}y = \frac{\sin x}{x}$. $P = 2/x$, so $\mu = e^{2\ln x} = x^2$ (taking $x > 0$, since the initial point $x = \pi$ is positive). Multiplying the *original* equation by x gives the same thing directly:

$$x^2y' + 2xy = (x^2y)' = x\sin x$$

Integration by parts: $\int x\sin x\,dx = \sin x - x\cos x + C$. So $x^2y = \sin x - x\cos x + C$.

Apply $y(\pi) = 0$: $0 = 0 - \pi(-1) + C$, so $C = -\pi$.

$$\boxed{y = \frac{\sin x - x\cos x - \pi}{x^2}, \qquad 0 < x < \infty}$$

$P(x) = 2/x$ is discontinuous at $x = 0$, so the interval of validity is the one containing $x = \pi$.

Problem 5

$$\frac{dy}{dx} = \frac{x^2 + y^2}{xy}$$

Solution: homogeneous

Numerator and denominator are both degree 2, so $f(tx, ty) = f(x, y)$. Let $y = ux$, so $y' = u + xu'$:

$$u + xu' = \frac{1 + u^2}{u} = \frac{1}{u} + u \quad \Longrightarrow \quad xu' = \frac{1}{u}$$

Separable: $u\,du = dx/x$, so $\frac{1}{2}u^2 = \ln|x| + C$. Put $u = y/x$ back:

$$\boxed{y^2 = 2x^2 \ln|x| + cx^2}$$

Problem 6

$\frac{dy}{dx} = \frac{y^2 - 1}{x}$. Find the general solution and any singular solutions.

Solution: separable, with a singular solution

$\dfrac{dy}{y^2-1} = \dfrac{dx}{x}$, which is only allowed if $y \neq \pm 1$. Partial fractions:

$$\frac{1}{y^2-1} = \frac{1}{2}\left(\frac{1}{y-1} - \frac{1}{y+1}\right)$$

$$\frac{1}{2}\ln\left|\frac{y-1}{y+1}\right| = \ln|x| + C \quad \Longrightarrow \quad \frac{y-1}{y+1} = cx^2$$

$$\boxed{y = \frac{1+cx^2}{1-cx^2}}$$

Now check the constant solutions you divided out. $y = 1$ is the family with $c = 0$, so it is already included. $y = -1$ **is a singular solution**: no finite c produces it.

Problem 7

$xy\,dx + (2x^2 + 3y^2 - 6)\,dy = 0$

Solution: not exact, integrating factor $\mu(y) = y^3$

$M = xy$ and $N = 2x^2 + 3y^2 - 6$, so $M_y = x$ and $N_x = 4x$. Not exact.

$(M_y - N_x)/N = -3x/(2x^2 + 3y^2 - 6)$ depends on both variables, so no $\mu(x)$. $(N_x - M_y)/M = 3x/(xy) = 3/y$ depends on y only, so

$$\mu(y) = e^{\int 3/y\, dy} = y^3$$

The equation becomes $xy^4\, dx + (2x^2y^3 + 3y^5 - 6y^3)\, dy = 0$. Check: both partials are $4xy^3$, so it is exact now.

$$F = \int xy^4\, dx = \tfrac{1}{2}x^2y^4 + h(y), \qquad F_y = 2x^2y^3 + h'(y) \quad \Longrightarrow \quad h'(y) = 3y^5 - 6y^3$$

$h = \frac{1}{2}y^6 - \frac{3}{2}y^4$. Multiply through by 2:

$$\boxed{x^2y^4 + y^6 - 3y^4 = c}$$

Problem 8

$\frac{dy}{dx} = 2xy + x, \qquad y(0) = 1$. Solve it two different ways.

Solution: both separable and linear

As linear: $y' - 2xy = x$. $\mu = e^{-x^2}$:

$$\left(ye^{-x^2}\right)' = xe^{-x^2} \quad \Longrightarrow \quad ye^{-x^2} = -\tfrac{1}{2}e^{-x^2} + C \quad \Longrightarrow \quad y = -\tfrac{1}{2} + Ce^{x^2}$$

As separable: the right side factors as $x(2y+1)$, so

$$\int \frac{dy}{2y+1} = \int x\,dx \quad \Longrightarrow \quad \tfrac{1}{2}\ln|2y+1| = \tfrac{1}{2}x^2 + C \quad \Longrightarrow \quad 2y+1 = Ae^{x^2}$$

Same family. Apply $y(0) = 1$: $1 = -\frac{1}{2} + C$, so $C = \frac{3}{2}$.

$$\boxed{y = \frac{3e^{x^2} - 1}{2}, \qquad -\infty < x < \infty}$$

On the quiz, go with whichever route you will make fewer mistakes on. Here the separable route is shorter.